



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
PRE-SCORE
08-10-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S(TEAS)]

PAPER-1

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

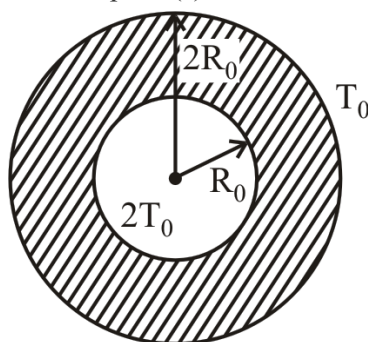
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. A spherical shell has variable thermal conductivity $K = K_0x$, where x is the distance from the center of the shell and K_0 is a positive constant. The inner and outer radii of the spherical shell are R_0 and $2R_0$ respectively as shown in the figure. Temperature of inner and outer surfaces are $2T_0$ and T_0 respectively. Then choose the **CORRECT** option(s).



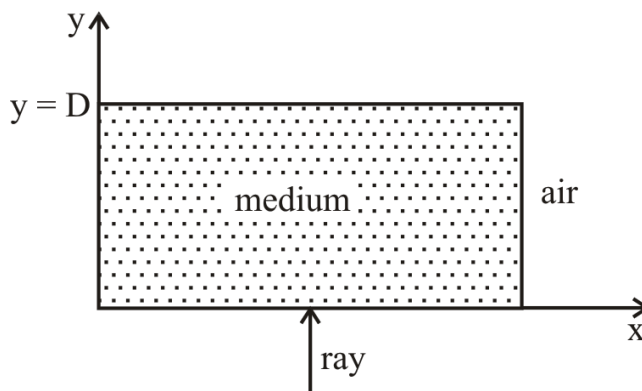
(A) Rate of heat flow is $\frac{8\pi K_0 R_0^2}{3} T_0$.

(B) Temperature at radial distance $x = \frac{3R_0}{2}$ is $\frac{3}{2} T_0$.

(C) Isothermal surfaces are spherical.

(D) The temperature at $x = R_0\sqrt{2}$ is $\frac{4T_0}{3}$.

2. A transparent slab is placed in x-y plane as shown in the figure. The refractive index of the medium is given as $\mu = \mu_0 \left[1 + \alpha \frac{y}{D} \right]$, where $\alpha = 2$. A ray is incident perpendicularly on the medium as shown in the diagram. Then :



- (A) Trajectory of the ray inside the medium is parabola.
 (B) Trajectory of the ray inside the medium is straight line.
 (C) The time taken by the ray to cross the medium is $\frac{2\mu_0 D}{c}$.
 (D) Equivalent optical path length of medium along trajectory of the ray is $2\mu_0 D$.
3. Bichromatic light of wavelength $\lambda_1 = 5000 \text{ \AA}$ and $\lambda_2 = 7000 \text{ \AA}$ are used in YDSE. Then :
- (A) 14th order maxima of λ_1 will coincide with 10th order maxima of λ_2 .
 (B) 21st order maxima of λ_2 will coincide with 15th order maxima of λ_1 .
 (C) 11th order minima of λ_1 will coincide with 8th order minima of λ_2 .
 (D) 3rd order minima of λ_1 will coincide with 4th order minima of λ_2 .

Space for Rough Work

4. A hypothetical radioactive substance decays according to $N = N_0 - kt^2$ where N is instantaneous number of active nuclei and N_0 is initial number of active nuclei and k is a positive constant of appropriate dimensions. Select correct option(s) [$t \leq \sqrt{\frac{N_0}{2k}}$]
- (A) Rate of decay when number of active nuclei becomes $\frac{N_0}{2}$ is $\sqrt{kN_0}$
- (B) Rate of decay when number of active nuclei becomes $\frac{N_0}{2}$ is $\sqrt{2kN_0}$.
- (C) Average life of radioactive sample is $\frac{2}{3} \sqrt{\frac{N_0}{k}}$.
- (D) Average life of radioactive sample is $\frac{1}{3} \sqrt{\frac{N_0}{k}}$.
5. A karate master strikes a pin board with a speed of 10 m/s which decreases to 1 m/s in a time interval of 0.002 s. The mass of hand and arm is 1 kg. The thickness of the pine board is 1 cm and volume 9.0 cm^3 . The breaking stress of the pine board is $3.36 \times 10^6 \text{ N/m}^2$. Then :
- (A) force on the board is 3500 N.
- (B) stress in the board due to the force is $5.0 \times 10^6 \text{ N/m}^2$.
- (C) the board will break, whatever be the mass of the board.
- (D) the board will not break if thickness is increased.

Space for Rough Work

6. One end of thin aluminium rod (cross-sectional Area = 10^{-6} m^2) is bult welded to the end of copper rod with same diameter. Both rods are very long and under uniform tension of 900 N. Given that

$$Y_{(\text{Cu})} = 1.6 \times 10^{11} \text{ Nm}^{-2} \quad Y_{(\text{Al})} = 0.9 \times 10^{11} \text{ N m}^{-2}$$

$$\rho_{(\text{Cu})} = 8.1 \times 10^3 \text{ Kg m}^{-3} \quad \rho_{(\text{Al})} = 2.5 \times 10^3 \text{ Kg m}^{-3}$$

- (A) The value of reflection factor $\frac{A_r}{A_i}$ for longitudinal pulse approaching the junction along copper rod is 0.15.
- (B) The value of reflection factor $\frac{A_r}{A_i}$ for longitudinal pulse approaching the junction along copper rod is approximately 0.3.
- (C) The value of reflection factor $\frac{A_r}{A_i}$ for transverse pulse approaching the junction along copper rod is 0.15.
- (D) The value of reflection factor $\frac{A_r}{A_i}$ for transverse pulse approaching the junction along copper rod is approximately 0.3.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. All the lenses used in list-I are thin, each of refractive index $\mu = 1.5$ and one of the surfaces is silvered. An object is shown moving in front of their silvered surface. Match with list-II suitably

List-I		List-II	
(I)		(P)	the image will coincide with the object at given instant
(II)		(Q)	magnitude of focal length ≥ 10 cm
(III)		(R)	magnitude of power ≥ 10 D
(IV)		(S)	speed of the image at the given instant ≥ 2 cm/s
		(T)	the image will move along the direction opposite to the direction of object's velocity

- (A) I $\rightarrow$ P,R,S,T; II $\rightarrow$ R,S,T; III $\rightarrow$ P,Q,T; IV $\rightarrow$ P,R,S,T
- (B) I $\rightarrow$ P,R,S; II $\rightarrow$ S,T; III $\rightarrow$ P,Q,T; IV $\rightarrow$ P,R,S
- (C) I $\rightarrow$ R,S,T; II $\rightarrow$ R,S,T; III $\rightarrow$ Q,S,T; IV $\rightarrow$ P,Q,R,S
- (D) I $\rightarrow$ P,R,S,T; II $\rightarrow$ P,Q,R,S,T; III $\rightarrow$ Q,R,S; IV $\rightarrow$ P,R,S,T

8. In List-I, certain thermodynamic process are given and in List-II, the value of physical quantities are given. Match the list-I with the list-II suitably.

ΔQ = amount of heat exchange,

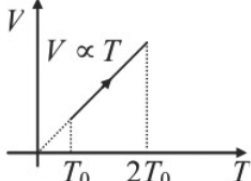
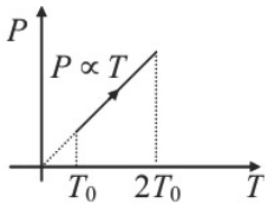
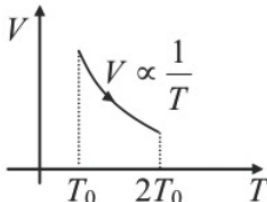
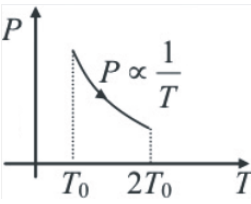
ΔU = change in internal energy,

ΔW = amount of pressure-volume work,

T = temperature of the gas,

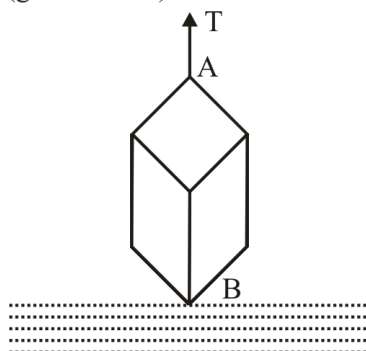
P = pressure of the gas,

ρ = density of the gas.

List-I		List-II	
(I)	1 mole of N_2 	(P)	$ \Delta Q \geq 2RT_0$
(II)	1 mole of He 	(Q)	$ \Delta U \geq \frac{3}{2}RT_0$
(III)	1 moles of a gaseous mixture having adiabatic index $\gamma = 1.5$ 	(R)	$ \Delta W \geq RT_0$
(IV)	1 moles of a gas having degree of freedom $f = 4$ 	(S)	ρ Vs T graph of the process is a straight line.
		(T)	P Vs ρ graph for the process is a parabola

- (A) I $\rightarrow$ P,Q,R; II $\rightarrow$ P,S,T; III $\rightarrow$ R,T; IV $\rightarrow$ Q,R,S,T
 (B) I $\rightarrow$ P,Q,R; II $\rightarrow$ Q,S; III $\rightarrow$ Q,R,T; IV $\rightarrow$ P,Q,R,T
 (C) I $\rightarrow$ P,Q,R,T; II $\rightarrow$ Q,S; III $\rightarrow$ P,Q,R,S,T; IV $\rightarrow$ Q,S
 (D) I $\rightarrow$ Q,R,T; II $\rightarrow$ P,Q,R; III $\rightarrow$ Q,S,T; IV $\rightarrow$ P,Q,R,S

9. A cube of volume 1 m^3 and specific gravity 2 is very slowly being lowered in to a lake with help of a massless string tied to one of its vertices as shown. Initially the vertex B is just touching the surface of water and finally, vertex A is just beneath the surface of water. For the process of immersion, match the lists: (Ignore change in KE of water), ($g = 10 \text{ ms}^{-2}$).



List-I		List-II	
(I)	Work done by gravity of water + cube system	(P)	$\frac{3\sqrt{3}}{2} \times 10^4 \text{ Joule}$
(II)	Work done by tension on the cube	(Q)	$\frac{-3\sqrt{3}}{2} \times 10^4 \text{ Joule}$
(III)	Work done by gravity on water	(R)	$\frac{\sqrt{3}}{2} \times 10^4 \text{ Joule}$
(IV)	Work done by gravity on the cube	(S)	$2\sqrt{3} \times 10^4 \text{ Joule}$
		(T)	$40\sqrt{3} \times 10^4 \text{ Joule}$

(A) $I \rightarrow S; II \rightarrow Q; III \rightarrow R; IV \rightarrow P$

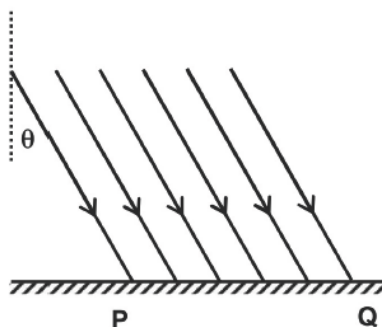
(B) $I \rightarrow P; II \rightarrow T; III \rightarrow S; IV \rightarrow R$

(C) $I \rightarrow Q; II \rightarrow R; III \rightarrow T; IV \rightarrow P$

(D) $I \rightarrow P; II \rightarrow Q; III \rightarrow R; IV \rightarrow S$

Space for Rough Work

10. A parallel beam of light is incident on a fixed surface PQ at an angle ' θ ' with the vertical as shown in the figure. The intensity of light is I and area of surface PQ is A . In List-I nature of reflection are given and in List-II force and generated radiation pressure are given. Then match List-I with suitable option List-II.
(Given that $I = 500 \text{ W/m}^2$, $A = 1.5 \text{ m}^2$, $\theta = 37^\circ$)



List-I		List-II	
(I)	If all the incident energy is absorbed by surface PQ then force on the surface and radiation pressure generated are respectively.	(P)	$3.2 \times 10^{-6} \text{ N}$, $2.13 \times 10^{-6} \text{ N/m}^2$
(II)	If all the incident energy is reflected by surface PQ then force on the surface and radiation pressure generated are respectively.	(Q)	$2.24 \times 10^{-6} \text{ N}$, $1.38 \times 10^{-6} \text{ N/m}^2$
(III)	If 30% incident energy is reflected by the surface then force on the surface and pressure generated are respectively.	(R)	$2.74 \times 10^{-6} \text{ N}$, $1.81 \times 10^{-6} \text{ N/m}^2$
(IV)	If 70% incident energy is reflected back then force on the surface and radiation pressure are respectively.	(S)	$2 \times 10^{-6} \text{ N}$, $1.06 \times 10^{-6} \text{ N/m}^2$
		(T)	$4.2 \times 10^{-6} \text{ N}$, $3.13 \times 10^{-6} \text{ N/m}^2$

(A) I $\rightarrow$ S; II $\rightarrow$ P; III $\rightarrow$ R; IV $\rightarrow$ Q

(B) I $\rightarrow$ R; II $\rightarrow$ Q; III $\rightarrow$ P; IV $\rightarrow$ S

(C) I $\rightarrow$ R; II $\rightarrow$ T; III $\rightarrow$ S; IV $\rightarrow$ Q

(D) I $\rightarrow$ S; II $\rightarrow$ P; III $\rightarrow$ Q; IV $\rightarrow$ R

Space for Rough Work

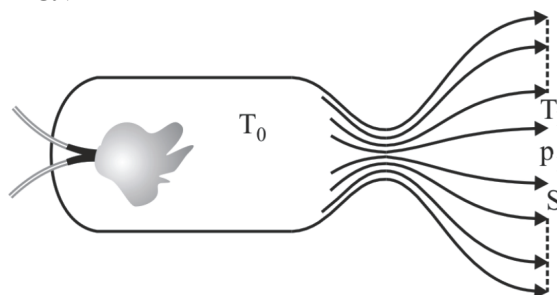
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

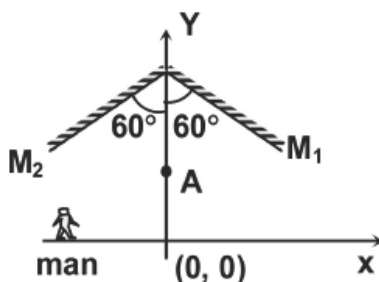
Zero Marks : 0 In all other cases.

1. In the nuclear reaction ${}_2\text{He}^4 + {}_4\text{Be}^9 \rightarrow {}_6\text{C}^{12} + {}_0\text{n}^1$, kinetic energy of the α -particle is 5.32 MeV. The Q-value of this reaction is $Q = 5.80$ MeV. The neutron goes perpendicular to the direction of the incoming α -particle. Find the kinetic energy (in MeV) of C^{12} produced.
2. In a vertical cylindrical vessel of base area $A = 60 \text{ cm}^2$ water is filled to a height $h = 10$ cm. If density and Bulk Modulus of water are $\rho = 1000 \text{ kg m}^{-3}$ and $B = 2 \times 10^9 \text{ Nm}^{-2}$ respectively, the elastic deformation energy of water in the vessel is found to be $n \times 10^{-7} \text{ J}$. Find n .
3. A liquid rocket engine is schematically shown in the figure below. Burning products (which is ideal monoatomic gas) are formed inside the combustion chamber at a high pressure and temperature and expand adiabatically while leaving the chamber through the nozzle. If the nozzle is correctly constructed (the neck diameter needs to correspond to the burning rate), exhaust gases continue adiabatic expansion after passing through the throat, up to the exit from the nozzle. The temperature of the exhaust gases inside the combustion chamber is T_0 ; at the exit, the cross-sectional area of the nozzle is S , temperature of exhaust gases is T_1 and pressure is p_1 . Find the thrust force F if $T_0 \gg T_1$, the effect of atmospheric pressure on the thrust can be neglected, and the kinetic energy of the gases inside the combustion chamber is negligible (as compared to the thermal energy). Take : $P_1 = 10^5 \text{ N/m}^2$, $S = 1 \text{ cm}^2$, $T_1 = 300 \text{ K}$, $T_0 = 1300 \text{ K}$.

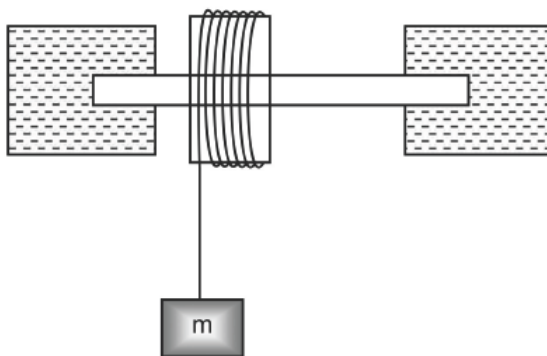


Space for Rough Work

4. Two large plane mirrors M_1 and M_2 are kept at an angle of 120° as shown in figure. The co-ordinate of intersection of plane mirror is $(0, 2 \text{ meter})$. A man moves along x-axis. Find the length (in meter) on x-axis for which man can see both the images of point A formed by mirrors simultaneously, (take $\sqrt{3} = 1.73$ and height of man is negligible)

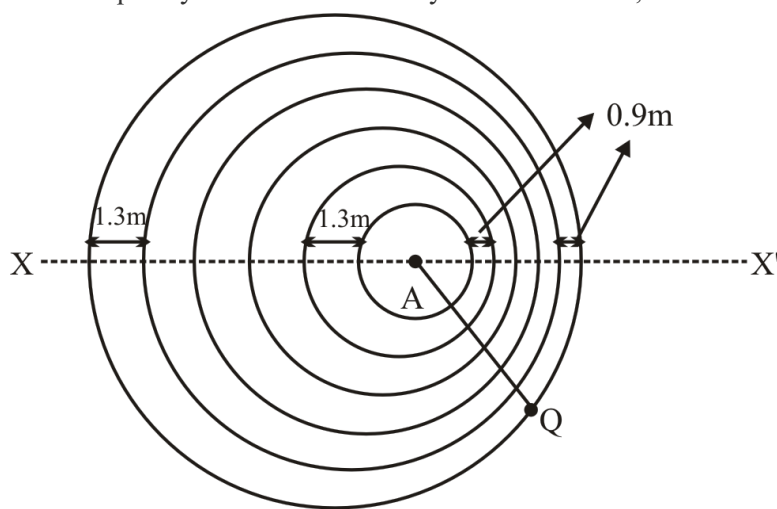


5. A block of mass $m = 2\text{kg}$ is attached with a string which is wrapped over a pulley of radius $R = 10 \text{ cm}$. The ends of the pulley are symmetrically supported by viscous liquids of equal amount. Heat capacity of each of the viscous liquids is $H = 100 \text{ J/}^\circ\text{C}$. It is found that block is moving down with a uniform velocity, $v_0 = 2.5 \text{ m/s}$. Find the rate of increase of temperature of the viscous liquids in $^\circ\text{C/s}$. Assume no loss of heat to the surrounding and the axle of the pulley does not take any heat. (Take $g = 10 \text{ m/s}^2$)



Space for Rough Work

6. A spherical soap bubble (surface tension = S) encloses n moles of a monoatomic ideal gas. The gas is heated slowly so that the surface area of the bubble increases by $\frac{3nR}{2S}$ per unit increment in temperature. The specific heat for this process is $\frac{xR}{2}$. Find the value of x (where x is least integer).
7. Figure shows wave fronts coming from a source at equal interval of $\frac{10}{3}$ milli second. Frequency of wave emitted by source is 300 Hz. XX' is line along common diameter of wave fronts passing through 'A'. Distance between two consecutive wave fronts along line XX' is 0.9 m and 1.3 m in the right and left of point A respectively. At some instant source is at point A. Detector is placed at point 'Q'. AQ makes an angle 60° with line XX' . If frequency of wave received by detector is f Hz, then the value of f is



8. A hydrogen atom in n^{th} excited state reaches the ground state by emitting three photons of frequency in ratio 5 : 72 : 243. Wavelength corresponding to one of the transitions belongs to visible region. Find the value of ' n '.

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. For a hydrogen like atom, the wave function for an electron is

$$\psi = \frac{1}{81} \cdot \sqrt{\frac{2}{\pi a_0^2}} \cdot \sigma \cdot (2 - \sigma) \cdot \cos \theta$$

where $\sigma = \frac{2Zr}{na_0}$ and $a_0 = 0.529 \text{ \AA}$. Select the correct information(s) about the orbital from which this electron belongs :

- | | |
|----------------------------|------------------------------|
| (A) Angular nodes = 1 | (B) Radial nodes = 1 |
| (C) Nodal plane = XZ plane | (D) Total nodal surfaces = 1 |

Space for Rough Work

2. 0.1 M – H_2A solution ($K_{a1} = 10^{-6}$, $K_{a2} = 10^{-10}$) is titrated against 0.1M-KOH solution. The correct information(s) regarding the relative concentrations of H_2A , HA^- and A^{2-} in the solution at various pH during the titration is/are :
- (A) At pH = 3.0, $[\text{H}_2\text{A}]$ is maximum
 (B) At pH = 8.0, $[\text{HA}^-]$ is maximum
 (C) At pH = 11.0, $[\text{A}^{2-}]$ is maximum
 (D) At pH = 4.0, $[\text{A}^{2-}]$ is maximum
3. The correct information(s) related with schrodinger's wave equation for H-like atom is/are :
- (A) Far from the nucleus, all radial wave functions approach zero, exponentially.
 (B) For s-orbital, angular function is zero.
 (C) For s-orbital, radial function is zero at $r = 0$.
 (D) For any orbital, the radial probability distribution function is zero at $r = 0$.
4. The reaction : $\text{AgCl(s)} \rightleftharpoons \text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq})$, is in equilibrium. The correct information(s) about the effect of given changes on this equilibrium is/are :
- (A) Equilibrium shifts right on increasing the temperature
 (B) Equilibrium shifts right on adding water.
 (C) Equilibrium concentration of $\text{Ag}^+(\text{aq})$ increased on adding water.
 (D) On adding NaCl(s) , the equilibrium concentration of Ag^+ decreases.

Space for Rough Work

5. For the reaction : $\text{C}_2\text{H}_4(\text{g}) + \text{H}_2(\text{g}) \rightleftharpoons \text{C}_2\text{H}_6(\text{g})$, the equilibrium concentration of $\text{C}_2\text{H}_6(\text{g})$ may be increased by :
- (A) Increasing the temperature
 - (B) Increasing the pressure
 - (C) Adding $\text{C}_2\text{H}_4(\text{g})$ at constant volume
 - (D) Adding $\text{Ne}(\text{g})$ at constant volume
6. Select the correct information(s) during the titration of H_3PO_4 solution by NaOH solution (For H_3PO_4 : $K_{a1} = 10^{-5}$, $K_{a2} = 10^{-8}$, $K_{a3} = 10^{-11}$) :
- (A) Three different kind of buffer solutions will form
 - (B) Three different equivalent points may be observed, using suitable indicators.
 - (C) pH of solution continuously increases.
 - (D) At second equivalent point, the solution must be neutral.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. Some properties related with electron in a H-like atom is given in List-I and its dependence (directly proportional) on the radius of orbit for electron is given in List-II. Select the correct match :

List-I		List-II	
(P)	Speed of electron	(1)	r
(Q)	Angular momentum of electron	(2)	$\frac{1}{r}$
(R)	Kinetic energy of electron	(3)	$\sqrt{r}$
(S)	Centripetal force on electron	(4)	$\frac{1}{\sqrt{r}}$
		(5)	$\frac{1}{r^2}$

(A) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$

(B) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 5$

(C) $P \rightarrow 4; Q \rightarrow 4; R \rightarrow 5; S \rightarrow 5$

(D) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 5$

Space for Rough Work

8. 10.0 ml of 0.1 M - H_2CO_3 solution is titrated with 0.2M - NaOH solution. Match correctly the volume of NaOH solution added (in List-I) with the pH of resulting solution (in List-II)

[For H_2CO_3 : $K_{a1} = 10^{-7}$, $K_{a2} = 10^{-11}$]

List-I		List-II	
(P)	0 ml	(1)	4.0
(Q)	2.5 ml	(2)	9.0
(R)	5.0 ml	(3)	7.0
(S)	7.5 ml	(4)	11.0
		(5)	6.0

- (A) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$ (B) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 5; S \rightarrow 4$
 (C) $P \rightarrow 1; Q \rightarrow 5; R \rightarrow 3; S \rightarrow 2$ (D) $P \rightarrow 1; Q \rightarrow 5; R \rightarrow 2; S \rightarrow 4$
9. Some reactions at equilibrium are given in List-I and some changes to disturb equilibrium are given in List-II. Match correctly the changes which results increase in concentration of product :

List-I		List-II	
(P)	$\text{A(g)} \rightleftharpoons 2\text{B(g)}; \Delta H = +ve$	(1)	Increase in temperature at constant volume
(Q)	$\text{A(s)} \rightleftharpoons 2\text{B(g)}; \Delta H = +ve$	(2)	Increase in pressure at constant temperature
(R)	$2\text{A(g)} \rightleftharpoons \text{B(g)}; \Delta H = -ve$	(3)	Increase in volume of system at constant temperature
(S)	$\text{A(g)} \rightleftharpoons \text{B(g)}; \Delta H = -ve$	(4)	Addition of He(g) at constant pressure and temperature
		(5)	Addition of B(g) at constant volume and temperature

- (A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$ (B) $P \rightarrow 1,2,5; Q \rightarrow 1; R \rightarrow 2,5; S \rightarrow 5$
 (C) $P \rightarrow 1,2,5; Q \rightarrow 1,3; R \rightarrow 2,4; S \rightarrow 2,5$ (D) $P \rightarrow 1,2,5; Q \rightarrow 4,5; R \rightarrow 2,5; S \rightarrow 3,5$

Space for Rough Work

10. Some electronic transition in H-atom is given in List-I and the wavelength of electromagnetic radiation emitted out is given in List-II. Match the correct wavelength for the given transition. [Rydberg constant = R_H]

List-I		List-II	
(P)	$n = 4$ to $n = 1$	(1)	$\frac{36}{5R_H}$
(Q)	$n = 3$ to $n = 2$	(2)	$\frac{4}{3R_H}$
(R)	$n = 5$ to $n = 3$	(3)	$\frac{225}{16R_H}$
(S)	$n = 2$ to $n = 1$	(4)	$\frac{16}{15R_H}$
		(5)	$\frac{3}{4R_H}$

- (A) $P \rightarrow 4; Q \rightarrow 5; R \rightarrow 3; S \rightarrow 2$
 (B) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 5$
 (C) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 2$
 (D) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 5$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. The ionisation energy of a hydrogen like atom is 5.5 M J/mol. the frequency of EMR emitted out in the transition $n = 5$ to $n = 4$ in this atom is $x \times 10^{14}$ Hz. The value of '2x' is [$h = 6.6 \times 10^{-34}$ J.s, $N_A = 6 \times 10^{23}$].
2. The radial function for 1s-orbital of hydrogen like atom is

$$\psi(r) = Z \cdot \left(\frac{Z}{a_0} \right)^{3/2} \cdot e^{-\frac{Zr}{a_0}}$$
 where $a_0 = 0.53 \text{ \AA}$.
 The most probable distance of 1s-electron from the nucleus in Li^{2+} ion is $x \times 10^{-12}$ m. The value of 'x' is :
3. pH of 0.75 M – BOH solution is (K_b of BOH = 2×10^{-13} , $\log 2 = 0.30$)
4. In 500 ml aqueous solution containing 0.2 moles H_2S and 0.1 mole NaHS, 0.1 mole NaOH is added. The increase in pH of solution is unit. [$\log 2 = 0.30$]
 [For H_2S : $K_{a1} = 10^{-7}$, $K_{a2} = 10^{-14}$]
5. Solubility product of Hg_2Cl_2 is 4×10^{-18} . The minimum mass (in mg) of $\text{Hg}_2\text{Cl}_2(\text{s})$ needed to form 1000 L saturated solution is ($\text{Hg} = 200$, $\text{Cl} = 35.5$)
6. For the reaction : $\text{A}(\text{g}) + \text{B}(\text{g}) \rightleftharpoons \text{C}(\text{g})$, $K_C = 0.2 \text{ M}^{-1}$. In a vessel of 10 L capacity, 5 moles of $\text{C}(\text{g})$ is taken. Moles of $\text{A}(\text{g})$ at equilibrium is :
7. The dissociation constant of weak acid, HA, is 2.0×10^{-13} . The pH of 1.0 M – NaA solution is ($\log 2 = 0.30$)
8. The equilibrium partial pressures of $\text{N}_2\text{O}_5(\text{g})$, $\text{NO}_2(\text{g})$ and $\text{O}_2(\text{g})$ are 0.4 atm, 0.8 atm and 0.2 atm respectively. The value of K_p of reaction : $2\text{N}_2\text{O}_5(\text{g}) \rightleftharpoons 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$, is $x \times 10^{-1} \text{ atm}^3$. The value of 'x' is :

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. Which of the following condition must be satisfied so that two of the lines represented by the equation $ay^4 + by^3x + cy^2x^2 + dxy^3 + ax^4 = 0$ will bisect the angle between the other two?

(A) $b + d = 0$

(B) $bd = 1$

(C) $c + 4a = 0$

(D) $c + 6a = 0$

Space for Rough Work

2. The circles $x^2 + y^2 - 4x - 81 = 0$, $x^2 + y^2 + 24x - 81 = 0$ intersect each other at points A and B. A line through point A meet one circle at P and a parallel line through B meet the other circle at Q. Let mid points of PQ lies on circle 'E₁' and set of all straight line segments joining a pair of distinct points of 'E₁' and passing through (-1, 1) be 'F'. Let 'E₂' be set of midpoints of line segments in set 'F' then which of the following is/are true ?
- (A) The point (-5, 5) lies on E₁
- (B) The point (-8, -4) does not lie on E₁
- (C) The point (-5, 1) lies on E₂
- (D) The point $(-5/2, 5/2)$ lies on E₂
3. Let $C_1 \equiv x^2 + y^2 = 64$ and $C_2 \equiv x^2 + y^2 - 24x + 108 = 0$ intersect at point P and Q. A line through P meets the circle C₁ and C₂ at R and S respectively. If PS = PR, then
- (A) length of PR = $\sqrt{130}$
- (B) length of perpendicular from centre of C₁ to PR is $\sqrt{\frac{63}{2}}$
- (C) length of perpendicular from centre of C₂ to PR is $\sqrt{\frac{7}{2}}$
- (D) slope of PR = $\frac{1}{\sqrt{3}}$

Space for Rough Work

4. Consider two family of straight lines $F_1 : 2x - y - 9 + \lambda(3x + y - 21) = 0$ and $F_2 : 2x + y - 5 + \mu(4x - y - 1) = 0$ where λ and μ are parameters. If L_1, L_2 are two lines belonging to F_1 and F_2 respectively, then identify which of the following statement(s) is(are) correct ?
- (A) If L_1 and L_2 are parallel ,then maximum distance between them is 5
- (B) If L_1 and L_2 are parallel ,then maximum distance between them is 3
- (C) If L_1 and L_2 are perpendicular, then locus of their points of intersection is $(x - 1)(x - 6) + (y - 3)^2 = 0$
- (D) If L_1 and L_2 are perpendicular, then locus of their points of intersection is $x^2 + y^2 - 6x + 6y + 9 = 0$
5. Let A be the set containing value(s) of x which satisfy the equation $3\sin^2 x - \cos x = \sin x(3\cos x - 1)$, where $x \in [0, 9\pi]$, then mark the correct statement(s).
- (A) Number of elements in set A is 17
- (B) Number of elements in set A is 18
- (C) Sum of elements in set A is $\frac{297\pi}{4}$
- (D) Sum of elements in set A is $\frac{295\pi}{4}$
6. Let R be a relation from the set $\{1, 2, 3, \dots, 60\}$ to itself such that $R = \{(a, b) : b = pq, \text{ where } p, q \geq 3 \text{ are prime numbers}\}$. Then, the number of elements in R is :
- (A) 600 (B) 660
- (C) 540 (D) 720

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

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7.

	List-I	List-II	
(I)	Number of solution(s) of the equation $e^x + e^{-x} = 2 \sin x$ for $x \in \left[0, \frac{\pi}{2}\right]$, is	(P)	0
(II)	Number of solution(s) of the equation $x + y = \frac{2\pi}{3}$ and $\cos x + \cos y = \frac{3}{2}$ is	(Q)	1
(III)	Number of solution(s) of the equation $\cos x + 2\sin x = 1$, $x \in (0, 2\pi]$ is	(R)	2
(IV)	Number of solution(s) of the equation $ \tan x = \sin 2x$ in $(0, 2\pi]$ is	(S)	3
		(T)	4

(A) I $\rightarrow$ P; II $\rightarrow$ P; III $\rightarrow$ R; IV $\rightarrow$ T

(B) I $\rightarrow$ Q; II $\rightarrow$ Q; III $\rightarrow$ S; IV $\rightarrow$ S

(C) I $\rightarrow$ R; II $\rightarrow$ R; III $\rightarrow$ Q; IV $\rightarrow$ R

(D) I $\rightarrow$ P; II $\rightarrow$ S; III $\rightarrow$ P; IV $\rightarrow$ Q

Space for Rough Work

8.

	List-I	List-II	
(I)	If the distance between the pair of parallel lines $x^2 + 2xy + y^2 - 8\lambda x - 8\lambda y - 9\lambda^2 = 0$ is $25\sqrt{2}$ then $ \lambda $ equals	(P)	1
(II)	If the lines represented by the equation $\lambda(\lambda - 3)x^2 + 16xy + (\lambda + 1)y^2 = 0$ are perpendicular to each other, then λ equals	(Q)	2
(III)	The image of point A(1, 2) by the line mirror $y = x$ is the point B and the image of B by the line mirror $y = 0$ is C and the image of C by the line mirror $x + y = 0$ is the point D(α , β) then $(\alpha + \beta)$ equals	(R)	5
(IV)	Given $\triangle ABC$ with unequal sides. P is the set of all the points which are equidistant from B and C and Q is the set of all the points which are equidistant from AB and AC, then $n(P \cap Q)$ equals	(S)	3
		(T)	6

 (A) $I \rightarrow R; II \rightarrow P; III \rightarrow S; IV \rightarrow Q$

 (B) $I \rightarrow P; II \rightarrow R; III \rightarrow S; IV \rightarrow Q$

 (C) $I \rightarrow Q; II \rightarrow P; III \rightarrow S; IV \rightarrow R$

 (D) $I \rightarrow R; II \rightarrow P; III \rightarrow Q; IV \rightarrow S$

Space for Rough Work

9. $S \equiv x^2 + y^2 - 2x - 4y + 1 = 0$ and $S' \equiv x^2 + y^2 - 4x - 2y + 1 = 0$ are equations of two circles.

	List-I	List-II	
(I)	Sum of number of tangents from (1, 3) to S and number of tangents from (1,3) to S' is	(P)	1
(II)	If r is radius of circle passes through point of intersection of S and S' and whose normal is the straight line $2x + y - 6 = 0$, then the value of $\frac{r^2}{4}$ is	(Q)	2
(III)	If m is slope of common tangent of $S = 0$ and $S' = 0$ then $ m $ is	(R)	3
(IV)	If ℓ is length of common tangent of S and S' then the value of $2\ell^2$ is	(S)	4
		(T)	6

- (A) $I \rightarrow Q; II \rightarrow P; III \rightarrow Q; IV \rightarrow R$
 (B) $I \rightarrow Q; II \rightarrow Q; III \rightarrow P; IV \rightarrow S$
 (C) $I \rightarrow Q; II \rightarrow Q; III \rightarrow S; IV \rightarrow P$
 (D) $I \rightarrow P; II \rightarrow P; III \rightarrow S; IV \rightarrow Q$

Space for Rough Work

10.

	List-I	List-II	
(I)	Circle passing through distinct points $(t^2, 1), (1, t), (2, 2t+1) \forall t \in \mathbb{R}$ passes through a fixed point (a, b) , then $a + b$ is	(P)	1
(II)	Tangent AB, AC are drawn from A to $x^2 + y^2 - 2x + 4y + 1 = 0$ meeting circle at B and C. If equation of circumcircle of ΔABC is $x^2 + y^2 - x + y - 2 = 0$ such that A is (a, b) , then $a + b$ is	(Q)	3
(III)	The value of $\left(16 \cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{3\pi}{7} \right)$ is	(R)	5
(IV)	Radius of a circle which is orthogonal to line pair $xy - 3x + 2y - 6 = 0$ and touches the line $x - 2y + 3 = 0$ is r then r^2 is	(S)	2
		(T)	8

(A) $I \rightarrow Q; II \rightarrow P; III \rightarrow S; IV \rightarrow R$

(B) $I \rightarrow Q; II \rightarrow R; III \rightarrow S; IV \rightarrow P$

(C) $I \rightarrow P; II \rightarrow Q; III \rightarrow S; IV \rightarrow R$

(D) $I \rightarrow Q; II \rightarrow P; III \rightarrow R; IV \rightarrow S$

Space for Rough Work

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- Answer to each question will be evaluated according to the following marking scheme:

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1. A straight line cuts the x-axis at point A(1,0), and y-axis at point B, such that $\angle OAB = \alpha \left(\alpha > \frac{\pi}{4} \right)$. C is middle point of AB and B' is a mirror image of point B with respect to line OC and C' is a mirror image of point C with respect to line BB', then the ratio of the areas of triangles ABB' and BB'C' is equal to
2. Two circles of radius 1 and 4 touch each other externally. Another circle of radius r touches both circles externally and also one direct common tangent of the two circles. Then $\left[\frac{1}{r} \right] = \underline{\hspace{2cm}}$
(Where $[\cdot]$ denotes geatest integer function)
3. The value of $\tan 1^\circ \cdot \tan 2^\circ + \tan 2^\circ \cdot \tan 3^\circ + \dots + \tan 88^\circ \cdot \tan 89^\circ$ is equal to $\cot^2 1^\circ - n$, then the value of n^2 is $\underline{\hspace{2cm}}$

Space for Rough Work

4. Let a variable line $\lambda(x - 2y + 3) + \mu(x + y - 6) = 0$, $\lambda, \mu \in \mathbb{R}$ divides a circle $S = 0$ into two regions of equal area. The circle $S = 0$ passes through the points on co-ordinate axes where one of the lines cuts off equal non zero intercept on the co-ordinate axes. If area of circle is $4k\pi$, then value of k is
5. The mean and standard deviation of 50 observations are 15 and 2 respectively. It was found that one incorrect observation was taken such that the sum of correct and incorrect observations is 70. If the correct mean is 16, then the correct variance is equal to :
6. Let $x_1, x_2, x_3, \dots, x_{20}$ be in geometric progression with $x_1 = 3$ and the common ratio $\frac{1}{2}$. A new data is constructed replacing each x_i by $(x_i - i)^2$. If $\bar{x}$ is the mean of new data, then the greatest integer less than or equal to $\bar{x}$ is
7. Let $A = \{1, 2, 3, 4, 5, 6, 7\}$ and $B = \{3, 6, 7, 9\}$. Then the number of elements in the set $\{C \subseteq A : C \cap B \neq \emptyset\}$ is
8. Let $S = \{4, 6, 9\}$ and $T = \{9, 10, 11, \dots, 1000\}$. If $A = \{a_1 + a_2 + \dots + a_k : k \in \mathbb{N}, a_1, a_2, a_3, \dots, a_k \in S\}$, then the sum of all the elements in the set $T - A$ is equal to

Space for Rough Work

Space for Rough Work

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NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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